

TUCUMCARI, NM, USA

WMO: 723676

Lat: 35.182N

Lon: 103.603W

Elev: 4065

StdP: 12.66

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 23048

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.0	(c) 16.8	(d) -2.3	(e) 5.6	(f) 42.4	(g) 1.1	(h) 6.7	(i) 41.3	(j) 34.8	(k) 57.2	(l) 31.7	(m) 55.8	(n) 6.3	(o) 60	(p) 0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	101.0	63.3	98.7	64.2	96.3	64.2	69.5	87.3	68.7	86.5	67.9	85.8	13.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	109.6	72.9	64.3	105.0	72.6	63.2	101.0	72.5	36.1	87.0	35.2	86.4	34.4	84.8	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.3	26.7	23.8	1.8	105.3	6.9	2.1	-3.2	106.9	-7.3	108.1	-11.2	109.3	-16.2	110.8
			0.9	72.2	6.5	3.3	-3.7	74.6	-7.5	76.6	-11.1	78.5	-15.8	80.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	38.9	42.3	50.7	58.8	67.2	78.1	81.0	79.0	72.0	59.7	47.9	39.0		
	DBStd	17.25	9.50	10.26	9.57	8.49	8.20	5.93	4.59	4.98	6.38	8.68	9.61	9.91		
	HDD50	1270	358	250	107	23	4	0	0	0	0	22	148	358		
	HDD65	3724	809	637	446	215	72	1	0	0	18	206	513	807		
	CDD50	4789	14	33	130	288	539	844	961	898	659	321	86	16		
	CDD65	1767	0	0	4	30	141	395	496	433	226	41	1	0		
	CDH74	22510	1	29	240	760	2145	5098	6151	4852	2494	660	78	2		
	CDH80	11912	0	3	51	237	1015	2947	3578	2681	1167	226	7	0		
(14) Wind	WSAvg	10.7	10.0	11.0	11.6	12.9	11.6	11.5	9.8	9.1	9.8	10.0	10.5	10.2		
(15) Precipitation	PrecAvg	14.8	0.2	0.4	0.4	0.9	1.5	1.7	3.3	2.7	1.7	1.0	0.6	0.3		
	PrecMax	27.3	1.2	1.3	2.0	3.4	5.0	4.3	6.9	7.0	5.0	4.4	2.9	1.1		
	PrecMin	8.8	0.0	0.0	0.0	0.0	0.2	0.1	0.4	0.4	0.2	0.0	0.0	0.0		
	PrecStd	4.3	0.3	0.4	0.4	0.8	1.3	1.2	1.7	2.1	1.3	1.1	0.7	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.2	78.2	85.4	90.3	97.4	105.1	104.5	101.9	97.4	90.6	80.6	72.2		
		MCWB	47.1	50.4	52.3	54.4	59.6	61.5	64.1	64.5	63.2	58.6	51.9	47.8		
	2%	DB	66.7	72.3	80.6	86.0	93.6	101.4	100.5	99.0	94.2	86.0	76.1	66.2		
		MCWB	45.3	47.3	50.7	53.0	58.0	60.9	65.5	65.6	62.6	57.8	50.7	45.7		
	5%	DB	62.0	67.1	75.7	82.3	90.3	98.7	98.2	96.5	91.1	81.8	71.1	61.5		
		MCWB	43.4	45.2	49.3	52.5	57.2	61.5	65.8	65.9	62.7	56.6	49.3	43.4		
	10%	DB	56.6	61.8	70.6	78.6	86.3	94.9	95.7	93.3	87.7	76.6	65.7	56.5		
		MCWB	40.9	43.0	47.4	51.8	56.9	62.5	65.7	65.9	62.3	55.3	47.2	41.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	51.4	56.4	59.4	66.2	70.2	71.0	71.1	68.9	66.6	56.4	50.1		
		MCDB	66.5	72.2	72.0	72.8	82.9	86.7	88.4	87.4	84.9	76.5	68.6	65.3		
	2%	WB	46.4	48.8	52.9	57.4	63.7	68.3	69.4	69.5	67.3	63.2	53.2	47.1		
		MCDB	63.9	68.6	72.9	73.1	81.0	86.5	88.3	86.8	81.6	72.5	68.8	64.4		
	5%	WB	44.1	46.4	50.7	55.4	61.9	66.9	68.6	68.6	66.1	60.1	50.8	44.5		
		MCDB	60.1	64.5	71.8	74.0	78.9	85.9	87.5	86.0	80.8	73.0	67.6	59.6		
	10%	WB	41.6	44.0	48.8	53.3	59.9	65.8	67.9	67.8	64.9	57.7	48.5	41.9		
		MCDB	55.5	60.7	68.7	74.1	76.4	84.8	86.8	85.3	79.9	72.8	63.8	55.6		
(35) Mean Daily Temperature Range	MDBR		28.5	29.5	30.8	30.9	29.9	30.0	27.3	26.6	26.9	27.9	28.3	26.5		
	5% DB	MCDBR	35.2	37.4	37.9	35.8	34.9	33.8	30.5	30.2	31.0	33.9	35.1	33.8		
		MCWBR	18.5	18.3	16.8	14.8	13.6	10.6	8.2	8.0	9.8	13.4	16.2	18.1		
	5% WB	MCDBR	33.0	34.4	34.2	30.3	29.9	28.9	27.1	26.7	25.5	26.0	30.7	31.7		
		MCWBR	17.9	17.5	15.9	14.5	12.8	9.9	7.6	7.7	8.9	12.1	14.9	17.7		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.257	0.282	0.310	0.323	0.361	0.380	0.363	0.331	0.292	0.266	0.253		
	taud		2.606	2.594	2.521	2.437	2.457	2.374	2.378	2.431	2.470	2.566	2.571	2.595		
	Ebn at Noon		311	319	316	309	304	291	285	289	295	301	301	303		
	Edh at Noon		23	26	31	36	36	39	38	36	32	27	24	22		
(44) All-Sky Solar Radiation	RadAvg		1031	1324	1724	2128	2370	2495	2367	2141	1829	1418	1098	894		
	RadStd		76	103	123	117	130	104	97	96	93	128	83	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	N/A	+1.60	N/A	N/A	N/A	N/A	+355	+241
(47) Regional (11 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+88

Nomenclature: See separate page